Two-Time Scale Control of Flexible Joint Robots With an Improved Slow Model

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***Abstract*—This paper presents a novel multivariable two-time scale control method for flexible joint robots. The fundamental limitations of the conventional two-time scale control are clearly explained based on a singular perturbation analysis with a linearized dynamic model. Unlike the conventional methods, the bandwidth of the slow control can be made high and the derivatives of the rigid control are not required. The closed-form solution of a slow model that is more accurate than the rigid model is presented. It is shown that the modeling error of the proposed two-time scale control can be made arbitrarily small by the control gain only. A real-world experiment with a six degrees-of-freedom commercial industrial robot was performed to validate the presented approach.**

***Index Terms*—Flexible joint robots, industrial robots, two-time scale control.**

I. Introduction

ONE of the trends in industrial robotics today is the lightweight design of the robot manipulator. In traditional robotic applications such as handling, welding, and machining, a lightweight design provides more opportunity to reduce the robot's cost as it permits one to use smaller actuators [1]. A lightweight design is also an important consideration for safety requirement as the robots are expected to work more closely with humans in the near future. However, as the robot is lighter, the control problem is more complex. In particular, the effect of joint elasticity can become more significant. If the joint elasticity is considered for control design, the dynamic model is more complex than a rigid case as the number of generalized coordinates is doubled. In addition, extra sensors on the link side or joint torque sensors are required to obtain high motion performance as a typical industrial robot has measurement only on the motor side [2].

Sweet and Good [3] performed a pioneering work on the importance of considering the joint elasticity for control design. Since then, control problems for flexible joint robots have been studied extensively. Amongst others, a method of full-state control has received much interest as it is theoretically an ideal control. That is, an arbitrary closed-loop performance can be achieved by full-state feedback. De Luca and Lucibello [4] showed that a general flexible joint dynamic model can only be feedback linearizable by dynamic state feedback. Spong [5] proposed a simplified model that can be feedback linearizable by static state feedback. Brogliato *et al*. [6] compared decoupling control, backstepping-based control, and passivity-based control. One common aspect of these full-state control methods is that they require feedback up to jerk, or the third derivative of the link position. It turned out that estimation of a high order state is very challenging in reality due to model uncertainty and measurement noise [7].

In the meantime, some researchers have suggested control methods based on low order states. Tomei [8] proved that the flexible joint dynamic model can achieve global stability with only position and velocity feedback on the motor side, i.e., partial-state feedback. The motor controller has been improved in several ways such as smooth velocity feedback [9], feedforward control [10], and gravity compensation [11]. The motor controller can be more welcome in industry as it can be easily implemented in a typical industrial robot due to sensor requirement and smoother input (feedback with low-order states). The price to pay, however, is the unsatisfactory performance on the link side due to the partial-state feedback [12].

Control methods developed based on a singular perturbation technique, denoted as *the singular perturbation control (SPC)* in this document, has also received much attention as it allows one to design full-state control with only position and velocity. In this method, the flexible joint dynamic model is split into two reduced models separated in different time scales assuming that the perturbation parameter, ε (the inverse of square root of joint stiffness), is sufficiently small. That is, the deflection dynamics or, the boundary layer represents the fast system, $\varepsilon\dot{\mathbf{z}} = \mathbf{g}(\mathbf{x}, \mathbf{z})$. The rigid dynamics is described as the quasi steady state, or the slow system, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_{\varepsilon=0})$. Composite control law is then applied to each subsystem, typically with velocity control for the fast control, $\mathbf{u}_f$, and a rigid control for the slow control, $\mathbf{u}_s = \mathbf{u}_{\varepsilon=0}$.

Spong *et al*. [13] proposed a corrective control, $\mathbf{u}_s = \mathbf{u}_0 + \varepsilon\mathbf{u}_1 + \varepsilon^2\mathbf{u}_2 + \cdots$, to improve the accuracy of the slow control. In [14] and [15], proportional-derivative (PD) control was suggested to increase the bandwidth of the fast system. Ghorbel and Spong [16] developed a condition of the perturbation parameter to guarantee stability. Taghirad and Khosravi [17] proposed a corrective control with integral action. Some real-world experiments were carried out to show that the composite control is more effective than a rigid control [18]–[20]. The SPC has also been applied to robots with flexible links [21], [22].

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In spite of this extensive research, there are two important questions that have not been answered properly. The two-time scale control assumes that the perturbation parameter is very small, $\varepsilon \approx 0$. The boundary layer can be made stable as long as a sufficient damping is added in the fast system, which can be easily implemented by velocity control or PD control. Therefore, the problem is simplified to controlling the slow system on the invariant manifold, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, \varepsilon))$, which is only a second-order reduced model. Unfortunately, the closed-form solution of the exact slow model is not available in the literature. Therefore, a rigid model, which is a simplified slow model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, 0))$, is often used for slow control. The first question is that it is not clear whether this slow model is accurate enough to represent the exact slow model as the modeling error of this model is $O(\varepsilon)$. It will be shown that this error is, in fact, quite large even for a typical industrial robot with a large joint stiffness. The second question is related to whether it is possible to reduce the modeling error of the slow model without increasing the joint stiffness. This question becomes critical when it is difficult to redesign the joint stiffness.

The contributions of this paper are as follows. First of all, the accuracy of the slow model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, 0))$, is analyzed. Since the exact model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, \varepsilon))$, is not available, the model is compared with a full model in terms of the closed-loop performance. More specifically, the slow control is designed such that the poles of the slow model are located in desired locations while the boundary layer remains strictly stable. Then, the composite control law is applied to the full model. If the perturbation parameter approaches zero, the slow poles of the full model must be placed near the desired locations. Therefore, these slow poles of the full model will be farther from the desired ones if the modeling error increases. Some researchers have suggested using corrective control for the slow model such that corrective terms, $\varepsilon \mathbf{u}_1 + \varepsilon^2 \mathbf{u}_2 + \cdots$, are added to the slow control, $\mathbf{u}_0$, that is designed on $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, 0))$. One major problem of this approach is that the corrective control involves derivatives of the rigid model, $\boldsymbol{\varphi}(\mathbf{u}_0(\mathbf{x}), 0)$. Since the rigid control, $\mathbf{u}_0$, is typically designed with PD control, the corrective control must contain high order states such as acceleration and jerk, thus defeating the original purpose of avoiding dependence on estimating high order states. It will be shown that a certain design of corrective control, e.g., [5], [19], have almost no effect because it can only compensate for the modeling error, $O(\varepsilon^2)$, not $O(\varepsilon)$.

Second, the closed-form solution of a more accurate slow model is presented. The model is in the form of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon \boldsymbol{\varphi}_1)$. Therefore, the modeling error of the proposed slow model is only $O(\varepsilon^2)$. The model is derived by explicit computation of the manifold condition. As a result, the slow control designed on the proposed slow model does not require a corrective term to compensate for the modeling error larger than $O(\varepsilon^2)$.

The control analysis and design in this paper is developed based on a linearized flexible joint dynamic model. Nonetheless, the presented study can be easily applied to a nonlinear model if feedforward control is applied since in many cases the nonlinearity can be effectively cancelled by feedforward control [10], [23].

The paper is organized as follows. Section II introduces the dynamic model that is considered in the paper. In Section III, the limitations of the conventional SPC are discussed, especially the slow control. Section IV obtains a new slow model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon \boldsymbol{\varphi}_1)$. In Section V, the slow models discussed in Sections III and IV are compared with an exemplar flexible joint robot. In Section VI, simulation and experimental results based on a six degrees-of-freedom (DOF) industrial robot are presented to validate our approach. Finally, concluding remarks are provided.

II. Dynamic Model

The dynamic model that is considered *the simplified flexible joint model* [5] and is denoted as follows:

$$\mathbf{M}_r(\mathbf{q}_r)\,\ddot{\mathbf{q}}_r + \mathbf{C}_r(\mathbf{q}_r, \dot{\mathbf{q}}_r) + \mathbf{K}(\mathbf{q}_r - \mathbf{r}^{-1}\mathbf{q}_m) = \mathbf{0} \ \in \ \Re^n \tag{1a}$$

$$\mathbf{M}_m\,\ddot{\mathbf{q}}_m + \mathbf{r}^{-1}\mathbf{K}(\mathbf{r}^{-1}\mathbf{q}_m - \mathbf{q}_r) = \boldsymbol{\tau}_m \ \in \ \Re^n \tag{1b}$$

where $\mathbf{q}_r$ and $\mathbf{q}_m$ represent the positions on the link and motor side, respectively. $\mathbf{M}_r$ is the link inertia (symmetric and positive definite). $\mathbf{C}_r$ includes the Coriolis and centrifugal effect and gravity. $\mathbf{K}$, $\mathbf{r}$, and $\mathbf{M}_m$ are constant diagonal matrices, representing joint stiffness, reduction ratio, and motor inertia, respectively. $\boldsymbol{\tau}_m$ is the motor torque.

The linear model for (1) can be written as follows:

$$\mathbf{M}_r\,\ddot{\mathbf{q}}_r + \mathbf{K}(\mathbf{q}_r - \mathbf{r}^{-1}\mathbf{q}_m) = \mathbf{0} \ \in \ \Re^n \tag{2a}$$

$$\mathbf{M}_m\,\ddot{\mathbf{q}}_m + \mathbf{r}^{-1}\mathbf{K}(\mathbf{r}^{-1}\mathbf{q}_m - \mathbf{q}_r) = \mathbf{u} \ \in \ \Re^n \tag{2b}$$

where $\mathbf{u}$ is the input of the linear system (2). It is assumed that the link inertia, $\mathbf{M}_r$, is constant. Therefore, the system (2) is linear with constant coefficients.

By defining $\boldsymbol{\mu} \triangleq \mathbf{K}(\mathbf{r}^{-1}\mathbf{q}_m - \mathbf{q}_r)$, the system (2) can be written as a singularly perturbed form,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{z}) = \mathbf{A}\mathbf{x} + \mathbf{B}\,\mathbf{z} \ \in \ \Re^{2n} \tag{3a}$$

$$\varepsilon\dot{\mathbf{z}} = \mathbf{g}(\mathbf{x}, \mathbf{z}) = \mathbf{C}\,\mathbf{z} + \mathbf{D}\,\mathbf{v} \ \in \ \Re^{2n} \tag{3b}$$

where $\mathbf{x} = [\,\mathbf{q}_r \ \dot{\mathbf{q}}_r\,]^T$: slow variable, $\mathbf{z} = [\,\boldsymbol{\mu} \ \varepsilon\dot{\boldsymbol{\mu}}\,]^T$: fast variable,

$$\mathbf{A} \triangleq \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{0} & \mathbf{0} \end{bmatrix},\ \mathbf{B} \triangleq \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{M}_r^{-1} & \mathbf{0} \end{bmatrix},$$

$$\mathbf{C} \triangleq \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\bar{\mathbf{K}}(\mathbf{J}^{-1} + \mathbf{M}_r^{-1}) & \mathbf{0} \end{bmatrix},\ \mathbf{D} \triangleq \begin{bmatrix} \mathbf{0} \\ -\bar{\mathbf{K}}\mathbf{J}^{-1} \end{bmatrix},$$

$$\mathbf{v} \triangleq \mathbf{r}\mathbf{u},\ \mathbf{J} \triangleq \mathbf{r}^2\mathbf{M}_m,\ \bar{\mathbf{K}} = \varepsilon^2\mathbf{K} = O(1),$$

and $0 < \varepsilon \ll 1$: the perturbation parameter.

The boundary layer is defined by

$$\boldsymbol{\eta}' = \mathbf{z}' - \varepsilon\dot{\boldsymbol{\varphi}}(\mathbf{x},\ \mathbf{v}_s(\mathbf{x})) \tag{4}$$

where $\boldsymbol{\eta}' \triangleq \frac{d}{d\tau}\boldsymbol{\eta}$, $d\tau = \frac{1}{\varepsilon}dt$. M_ε : $\mathbf{z} = \boldsymbol{\varphi}(\mathbf{x}, \varepsilon)$ is the invariant manifold. $\boldsymbol{\eta} \triangleq \mathbf{z} - \boldsymbol{\varphi}(\mathbf{x}, \varepsilon)$ is the deviation of the fast variable, $\mathbf{z}$ from M_ε and $\mathbf{v}_s$ is the slow control. If the boundary

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layer is strictly stable, i.e., $\dot{\boldsymbol{\eta}} = \mathbf{0}$, $\boldsymbol{\eta} = \mathbf{0}$, then the solution of (2) satisfies $\mathbf{z} = \boldsymbol{\varphi}(\mathbf{x}, \varepsilon)$.

The composite control is defined as

$$\mathbf{v} = \mathbf{v}_f(\mathbf{z}) + \mathbf{v}_s(\mathbf{x}) \tag{5}$$

where $\mathbf{v}_f(\mathbf{z})$: fast control.

III. Conventional Control

It is known that the boundary layer (4) can be stable with velocity control, namely,

$$\mathbf{v}_f = \mathbf{L}_d \mathbf{z}_2 \tag{6}$$

where $\mathbf{L}_d$ is a constant diagonal matrix. In this case, the equilibrium manifold, $M_0 : \bar{\mathbf{z}} = \boldsymbol{\varphi}(\mathbf{x}, 0) = \boldsymbol{\varphi}_0$, is given by

$$\boldsymbol{\varphi}_0 = -\mathbf{C}^{-1}\mathbf{D}\mathbf{v}_s = \begin{bmatrix} -\mathbf{M}_r(\mathbf{J}+\mathbf{M}_r)^{-1}\mathbf{v}_s \\ \mathbf{0} \end{bmatrix}. \tag{7}$$

Therefore, the slow model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, 0))$, becomes

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\boldsymbol{\varphi}_0 = \begin{bmatrix} \mathbf{x}_2 \\ -(\mathbf{J}+\mathbf{M}_r)^{-1}\mathbf{v}_s \end{bmatrix} \tag{8}$$

which is the rigid dynamics

$$(\mathbf{J}+\mathbf{M}_r)\,\dot{\mathbf{x}}_2 + \mathbf{v}_s = \mathbf{0}. \tag{9}$$

Therefore, the slow control is typically chosen as decoupling control

$$\mathbf{v}_s = \hat{\mathbf{K}}_s \mathbf{x} \tag{10}$$

where $\hat{\mathbf{K}}_s \triangleq (\mathbf{J}+\mathbf{M}_r)\mathbf{K}_s$, $\mathbf{K}_s = [\,\mathbf{K}_p \;\; \mathbf{K}_d\,]$. $\mathbf{K}_p$ and $\mathbf{K}_d$ are constant diagonal matrices.

The modeling error of (7) compared with the exact slow model is $O(\varepsilon)$. The corrective control aims for reducing this error by adding corrective terms to the slow input such that the slow model approximates the exact slow model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1 + \cdots) \approx \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, \varepsilon))$. The initial trial is to find the first-order approximation. To do so, choose $\boldsymbol{\varphi} = \boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1$ and $\mathbf{v}_s = \mathbf{v}_0 + \varepsilon\mathbf{v}_1$, then insert them into the manifold condition (3b) to obtain

$$\varepsilon\,(\dot{\boldsymbol{\varphi}}_0 + \varepsilon\dot{\boldsymbol{\varphi}}_1) = \mathbf{C}\,(\boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1) + \mathbf{D}\,(\mathbf{v}_0 + \varepsilon\mathbf{v}_1). \tag{11}$$

Since ε is not zero, (11) must satisfy

$$\mathbf{C}\boldsymbol{\varphi}_0 + \mathbf{D}\mathbf{v}_0 = \mathbf{0} \tag{12}$$

$$\mathbf{D}\mathbf{v}_1 + \mathbf{C}\boldsymbol{\varphi}_1 - \dot{\boldsymbol{\varphi}}_0 = \mathbf{0}. \tag{13}$$

The condition (12) suggests that $\mathbf{v}_0$ be designed on M_0 as (10), whereas (13) is expanded by the power series,

$$\mathbf{D}\mathbf{v}_1 = \frac{\partial \boldsymbol{\varphi}_0}{\partial \mathbf{x}}\mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0) - \mathbf{C}\left(\frac{\partial \mathbf{g}}{\partial \mathbf{z}}\right)^{-1}_{z=\varphi_0} \frac{\partial \boldsymbol{\varphi}_0}{\partial \mathbf{x}}\mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0). \tag{14}$$

Thus, $\mathbf{D}\mathbf{v}_1 = \mathbf{0}$ because $\frac{\partial \mathbf{g}}{\partial \mathbf{z}} = \mathbf{C}$ and it is nonsingular. However, this result implies

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0) = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1) \tag{15}$$

because

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1) \\ &= \mathbf{A}\mathbf{x} + \mathbf{B}\boldsymbol{\varphi}_0 + \varepsilon\mathbf{B}\mathbf{C}^{-1}\frac{\partial \boldsymbol{\varphi}_0}{\partial x}(\mathbf{A}\mathbf{x} + \mathbf{B}\boldsymbol{\varphi}_0) \\ &= \mathbf{A}\mathbf{x} + \mathbf{B}\boldsymbol{\varphi}_0 = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0) \;(\because\; \mathbf{B}\mathbf{C}^{-2}\mathbf{D} = \mathbf{0}). \end{aligned} \tag{16}$$

The term, $\boldsymbol{\varphi}_1$, is given by

$$\begin{aligned} \boldsymbol{\varphi}_1 &= \left(\frac{\partial \mathbf{g}}{\partial \mathbf{z}}\right)^{-1}_{\mathbf{z}=\boldsymbol{\varphi}_0} \frac{\partial \boldsymbol{\varphi}_0}{\partial \mathbf{x}}\mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0) \\ &= -\mathbf{C}^{-2}\mathbf{D}\hat{\mathbf{K}}_s\left(\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{D}\hat{\mathbf{K}}_s\right)\mathbf{x}. \end{aligned} \tag{17}$$

It can be easily verified that the following is satisfied:

$$C(\boldsymbol{\varphi}_0) \perp C(\boldsymbol{\varphi}_1) \tag{18}$$

where $C(\bullet)$ is the column space of a matrix $\bullet$. More specifically, the manifold is mapped to the slow model by $\mathbf{B}$ as shown in (8),

$$\mathbf{B}\boldsymbol{\varphi} = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{M}_r^{-1} & \mathbf{0} \end{bmatrix}\begin{bmatrix} \boldsymbol{\varphi}^1 \\ \boldsymbol{\varphi}^2 \end{bmatrix}. \tag{19}$$

Therefore, only the first n lows of $\boldsymbol{\varphi}$ will contribute to the slow model. The first n lows of $\boldsymbol{\varphi}_0$ are not zeros [see (7)] whereas they are all zeros in $\boldsymbol{\varphi}_1$. This suggests that the corrective control designed on $\boldsymbol{\varphi}_1$ cannot reach the equilibrium manifold, $\boldsymbol{\varphi}_0$, because the vector spaces spanned by these manifolds are orthogonal to each other.

Therefore, the following corrective control is pursued.

$$\mathbf{v}_s = \mathbf{v}_0 + \varepsilon^2\mathbf{v}_2. \tag{20}$$

Again, applying it to the manifold condition,

$$\varepsilon\,(\dot{\boldsymbol{\varphi}}_0 + \varepsilon\dot{\boldsymbol{\varphi}}_1) = \mathbf{C}\,(\boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1) + \mathbf{D}\left(\mathbf{v}_0 + \varepsilon^2\mathbf{v}_2\right). \tag{21}$$

The term with ε^2 satisfies

$$\begin{aligned} \mathbf{D}\mathbf{v}_2 = \mathbf{C}^{-1}\ddot{\boldsymbol{\varphi}}_0 &= \mathbf{C}^{-1}\frac{d}{dt}\left(\frac{\partial \boldsymbol{\varphi}_0}{\partial \mathbf{x}}(\mathbf{A}\mathbf{x} + \mathbf{B}\boldsymbol{\varphi}_0)\right) \\ &= \mathbf{C}^{-1}\frac{\partial \boldsymbol{\varphi}_0}{\partial \mathbf{x}}\left(\mathbf{A}\mathbf{x} + \mathbf{B}\frac{\partial \boldsymbol{\varphi}_0}{\partial \mathbf{x}}\right)(\mathbf{A}\mathbf{x} + \mathbf{B}\boldsymbol{\varphi}_0). \end{aligned} \tag{22}$$

Equivalently,

$$\mathbf{D}\mathbf{v}_2 = -\mathbf{C}^{-2}\mathbf{D}\hat{\mathbf{K}}_s\left(\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{D}\hat{\mathbf{K}}_s\right)^2\mathbf{x}. \tag{23}$$

Therefore,

$$\mathbf{v}_2 = \mathbf{J}\bar{\mathbf{K}}^{-1}\mathbf{M}_r\left[\mathbf{K}_d(\mathbf{K}_d{}^2 - 2\mathbf{K}_p)\dot{\mathbf{q}}_r + \mathbf{K}_p(\mathbf{K}_d{}^2 - \mathbf{K}_p)\mathbf{q}_r\right]. \tag{24}$$

In fact, the corrective control (20) transforms the rigid model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x},\ \boldsymbol{\varphi}_0)$, into $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x},\ \boldsymbol{\varphi}_0 + \varepsilon^2\boldsymbol{\varphi}_2)$. The term, $\boldsymbol{\varphi}_2$, can be approximated by the power series,

$$\begin{aligned} \boldsymbol{\varphi}_2 = &\left(\frac{\partial \mathbf{g}}{\partial \mathbf{z}}\right)^{-1}_{\mathbf{z}=\boldsymbol{\varphi}_0}\frac{\partial \boldsymbol{\varphi}_0}{\partial \mathbf{x}}\frac{\partial \mathbf{f}}{\partial \mathbf{z}}\Big|_{\mathbf{z}=\boldsymbol{\varphi}_0}\boldsymbol{\varphi}_1 \\ &+\frac{\partial \boldsymbol{\varphi}_1}{\partial \mathbf{x}}\mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0) - \frac{\partial^2 \mathbf{g}}{\partial \mathbf{z}^2}\Big|_{\mathbf{z}=\boldsymbol{\varphi}_0}\boldsymbol{\varphi}_1{}^2. \end{aligned} \tag{25}$$

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The partial derivative of $\boldsymbol{\varphi}_1$ is given by

$$\frac{\partial \boldsymbol{\varphi}_1}{\partial \mathbf{x}} = -\mathbf{C}^{-2}\mathbf{D}\hat{\mathbf{K}}_s\left(\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{D}\hat{\mathbf{K}}_s\right). \tag{26}$$

Finally, we obtain

$$\boldsymbol{\varphi}_2 = \left[\mathbf{C}^{-2}\mathbf{D}\hat{\mathbf{K}}_s\mathbf{B}\mathbf{C}^{-2}\mathbf{D}\hat{\mathbf{K}}_s - \mathbf{C}^{-3}\mathbf{D}\hat{\mathbf{K}}_s\left(\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{D}\hat{\mathbf{K}}_s\right)\right] \times \left(\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{D}\hat{\mathbf{K}}_s\right)\mathbf{x}. \tag{27}$$

From (17) and (23),

$$\boldsymbol{\varphi}_2 = \mathbf{C}^{-2}\mathbf{D}\hat{\mathbf{K}}_s\mathbf{B}\boldsymbol{\varphi}_1 + \mathbf{C}^{-1}\mathbf{D}\mathbf{v}_2. \tag{28}$$

Since it holds $\mathbf{B}\boldsymbol{\varphi}_1 = \mathbf{0}$ by (19),

$$\boldsymbol{\varphi}_2 = \mathbf{C}^{-1}\mathbf{D}\mathbf{v}_2. \tag{29}$$

Therefore,

$$\boldsymbol{\varphi} = \boldsymbol{\varphi}_0 + \varepsilon^2\boldsymbol{\varphi}_2 = -\mathbf{C}^{-1}\mathbf{D}\left(\mathbf{v}_0 + \varepsilon^2\mathbf{v}_2\right) = -\mathbf{C}^{-1}\mathbf{D}\mathbf{v}_s. \tag{30}$$

As proved above, the corrective control (20) is an equivalent implementation of the slow model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon^2\boldsymbol{\varphi}_2)$. However, the modeling error of this model is $O(\varepsilon)$, which may be large as the first order correction term, $\varepsilon\boldsymbol{\varphi}_1$, is not included in the model.

IV. Proposed Control

In this section, a more accurate slow model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1)$ is derived. The previous analysis indicates that the fundamental reason why the conventional method fails to obtain $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1)$ is the fact, $\mathbf{B}\mathbf{C}^{-2}\mathbf{D} = \mathbf{0}$ (16). However, it is possible to make this term nonzero by a fast control in the form of $\mathbf{v}_f = \mathbf{L}\mathbf{z}$ in (3b) such that the matrix $\mathbf{C}$ is modified as $\bar{\mathbf{C}} = \mathbf{C} + \mathbf{D}\mathbf{L}$. One can verify that $\mathbf{B}\bar{\mathbf{C}}^{-2}\mathbf{D} \neq \mathbf{0}$ is satisfied as long as damping is added to the fast system, e.g., $\mathbf{v}_f = \mathbf{L}_d\mathbf{z}_2$. However, the velocity control alone is not sufficient to achieve an arbitrary bandwidth of the fast system [14]. Instead of using PD control, one can consider decoupling control, namely,

$$\mathbf{v}_f = \bar{\mathbf{L}}_f\,\mathbf{z} \tag{31}$$

where $\bar{\mathbf{L}}_f = \mathbf{J}\bar{\mathbf{K}}^{-1}\mathbf{L}_f - [\,\mathbf{I} + \mathbf{J}\mathbf{M}_r^{-1}\ \mathbf{0}\,]$, $\mathbf{L}_f = [\,\mathbf{L}_p\ \mathbf{L}_d\,]$ such that the fast system becomes

$$\mathbf{z}' = \mathbf{L}_f\,\mathbf{z}. \tag{32}$$

With the fast control (31), the model (3) can still be written in a singularly perturbed form,

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{z} \tag{33a}$$

$$\varepsilon\dot{\mathbf{z}} = \bar{\mathbf{C}}\mathbf{z} + \mathbf{D}\mathbf{v}_s \tag{33b}$$

where $\bar{\mathbf{C}} = \left[\begin{smallmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{L}_p & -\mathbf{L}_d \end{smallmatrix}\right]$. As a result, the equilibrium manifold is modified as

$$\boldsymbol{\varphi}_0 = -\bar{\mathbf{C}}^{-1}\mathbf{D}\,\mathbf{v}_s. \tag{34}$$

The slow model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, 0))$, is obtained as

$$\mathbf{M}_r\ddot{\mathbf{q}}_r + \mathbf{L}_p^{-1}\bar{\mathbf{K}}\mathbf{J}^{-1}\mathbf{v}_s = \mathbf{0}. \tag{35}$$

Consider decoupling control for the slow control,

$$\mathbf{v}_s = \bar{\mathbf{K}}_s\mathbf{x} \tag{36}$$

where $\bar{\mathbf{K}}_s \triangleq \mathbf{J}\bar{\mathbf{K}}^{-1}\mathbf{L}_p\mathbf{M}_r\mathbf{K}_s$, such that the slow system becomes

$$\dot{\mathbf{x}} = \mathbf{K}_s\mathbf{x}. \tag{37}$$

Now let us obtain the slow model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x},\ \boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1)$. First of all, it is necessary to obtain

$$\boldsymbol{\varphi}_0 = -\bar{\mathbf{C}}^{-1}\mathbf{D}\,\mathbf{v}_s = \begin{bmatrix} -\mathbf{M}_r\mathbf{K}_p & -\mathbf{M}_r\mathbf{K}_d \\ \mathbf{0} & \mathbf{0} \end{bmatrix}\mathbf{x}. \tag{38}$$

Similar to (17), $\boldsymbol{\varphi}_1$ can be written as

$$\boldsymbol{\varphi}_1 = -\bar{\mathbf{C}}^{-2}\mathbf{D}\bar{\mathbf{K}}_s\left(\mathbf{A} - \mathbf{B}\bar{\mathbf{C}}^{-1}\mathbf{D}\bar{\mathbf{K}}_s\right)\mathbf{x} \tag{39}$$

where

$$\bar{\mathbf{C}}^{-2}\mathbf{D}\bar{\mathbf{K}}_s = \begin{bmatrix} -\mathbf{L}_p^{-1}\mathbf{L}_d\mathbf{M}_r\mathbf{K}_p & -\mathbf{L}_p^{-1}\mathbf{L}_d\mathbf{M}_r\mathbf{K}_d \\ \mathbf{M}_r\mathbf{K}_p & \mathbf{M}_r\mathbf{K}_d \end{bmatrix},$$

$$\mathbf{B}\bar{\mathbf{C}}^{-1}\mathbf{D}\bar{\mathbf{K}}_s = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{K}_p & \mathbf{K}_d \end{bmatrix},\ \bar{\mathbf{C}}^{-1} = \begin{bmatrix} -\mathbf{L}_p\mathbf{L}_d & -\mathbf{L}_p \\ \mathbf{I} & \mathbf{0} \end{bmatrix},$$

and $\bar{\mathbf{C}}^{-2} = \mathbf{L}_p\begin{bmatrix} \mathbf{L}_d\mathbf{L}_p\mathbf{L}_d - \mathbf{I} & \mathbf{L}_d\mathbf{L}_p \\ -\mathbf{L}_d & -\mathbf{I} \end{bmatrix}$.

Therefore,

$$\boldsymbol{\varphi}_1 = \begin{bmatrix} -\mathbf{L}_p^{-1}\mathbf{L}_d\mathbf{M}_r\mathbf{K}_d\mathbf{K}_p & \mathbf{L}_p^{-1}\mathbf{L}_d\mathbf{M}_r\mathbf{K}_p - \mathbf{L}_p^{-1}\mathbf{L}_d\mathbf{M}_r\mathbf{K}_d^2 \\ \mathbf{M}_r\mathbf{K}_d\mathbf{K}_p & -\mathbf{M}_r\mathbf{K}_p + \mathbf{M}_r\mathbf{K}_d^2 \end{bmatrix}\mathbf{x}. \tag{40}$$

The first-order approximation of the invariant manifold, $\boldsymbol{\varphi}(\mathbf{x}, \varepsilon)$, can be written in the form of

$$\boldsymbol{\varphi} = \boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1 = \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix}\mathbf{x}. \tag{41}$$

Thus, it can be written as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x},\ \boldsymbol{\varphi}_0 + \varepsilon\boldsymbol{\varphi}_1) = \begin{bmatrix} \mathbf{x}_2 \\ \mathbf{M}_r^{-1}\boldsymbol{\gamma}_{11}\mathbf{x}_1 + \mathbf{M}_r^{-1}\boldsymbol{\gamma}_{12}\mathbf{x}_2 \end{bmatrix}. \tag{42}$$

More explicitly,

$$\mathbf{M}_r\ddot{\mathbf{q}}_r + \left[\mathbf{M}_r\mathbf{K}_d + \varepsilon\mathbf{L}_p^{-1}\mathbf{L}_d\mathbf{M}_r\left(\mathbf{K}_d^2 - \mathbf{K}_p\right)\right]\dot{\mathbf{q}}_r + \left[\mathbf{M}_r\mathbf{K}_p + \varepsilon\mathbf{L}_p^{-1}\mathbf{L}_d\mathbf{M}_r\mathbf{K}_d\mathbf{K}_p\right]\mathbf{q}_r = \mathbf{0}. \tag{43}$$

V. Model Comparison

In this section, the modeling errors of the slow models discussed in the previous sections are investigated with an exemplar flexible joint robot. The robot is assumed to have a single link and it represents the first link (arm sweep) of a real commercial industrial robot described in Section VI. The values of the model parameters are, $J = 6.926\ \mathrm{kg/m^2}$, $J_r = 16.681\ \mathrm{kg/m^2}$ (link inertia), and $k = 100 \times 10^3\ \mathrm{Nm/rad}$. The natural frequency on the link side is approximately 12 Hz.

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For a single link robot, the slow models and the full model can be written as follows:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0) : \ddot{q}_r + k_d \dot{q}_r + k_p q_r = 0 \tag{44}$$

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon^2 \boldsymbol{\varphi}_2) : \ddot{q}_r + \gamma_{\varepsilon^2,d}\, \dot{q}_r + \gamma_{\varepsilon^2,p}\, q_r = 0 \tag{45}$$

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}_0 + \varepsilon \boldsymbol{\varphi}_1) : \ddot{q}_r + \gamma_{\varepsilon,d}\, \dot{q}_r + \gamma_{\varepsilon,p}\, q_r = 0 \tag{46}$$

full model:

$$\frac{\varepsilon^2}{{\omega_f}^2} \overset{(4)}{q}_r + \frac{\varepsilon 2\zeta}{\omega_f} \overset{(3)}{q}_r + \ddot{q}_r + k_d \dot{q}_r + k_p q_r = 0 \tag{47}$$

where

$$\gamma_{\varepsilon^2,d} \triangleq k_d \left[1 + \varepsilon^2 J_e \left(2k_p - {k_d}^2\right)\right],$$

$$\gamma_{\varepsilon^2,p} \triangleq k_p \left[1 + \varepsilon^2 J_e \left(k_p - {k_d}^2\right)\right],$$

$$\gamma_{\varepsilon,d} \triangleq k_d + \varepsilon \frac{l_d}{l_p}\left(k_d^2 - k_p\right),$$

$$\gamma_{\varepsilon,p} \triangleq k_p + \varepsilon \frac{l_d}{l_p} k_d k_p, \quad J_e \triangleq \frac{J J_r}{J + J_r}.$$

Let us assume that the desired performance of the fast system is given by

$$\mu'' + 2\zeta\omega_f \mu' + {\omega_f}^2 \mu = 0. \tag{48}$$

The fast systems in Sections III and IV can be written, respectively, as follows:

$$\mu'' + \frac{l_d}{J}\mu' + {J_e}^{-1}\mu = 0 \tag{49a}$$

$$\mu'' + l_d \mu' + l_p \mu = 0. \tag{49b}$$

The above fast systems can be in the form of (48) by choosing

$$l_d = 2\zeta J \omega_f, l_p = {\omega_f}^2, \omega_f = {J_e}^{-1/2}. \tag{50}$$

The full model (47) approximates the rigid model (44) as $\varepsilon \to 0$ or $\omega_f \to \infty$. Since the bandwidth of the fast system, ω_f, cannot be arbitrarily small due to (50), the modeling error heavily depends on ε in this case.

In Fig. 1, the four models are compared in the frequency domain assuming that the gain of the slow systems is selected according to

$$k_d = 2\zeta\omega_s, k_p = {\omega_s}^2. \tag{51}$$

The models (44) and (45) are very similar in all frequency range. The error of the model (45) is slightly smaller than the rigid model only at high frequency. These models have small errors only at low frequency inside their bandwidths. However, the errors increase as the models approach their upper limit bandwidths and these errors become more serious as the gain, ω_s, increases [see Fig. 1(b)]. This means that the gains of the slow systems must be kept small. This explains why the conventional SPC cannot be used for a time-optimal trajectory.

The error of the proposed slow model (46) is also not small. The error is large at high frequency. However, the error is much smaller than the models (44) and (45) near the upper limit bandwidth.

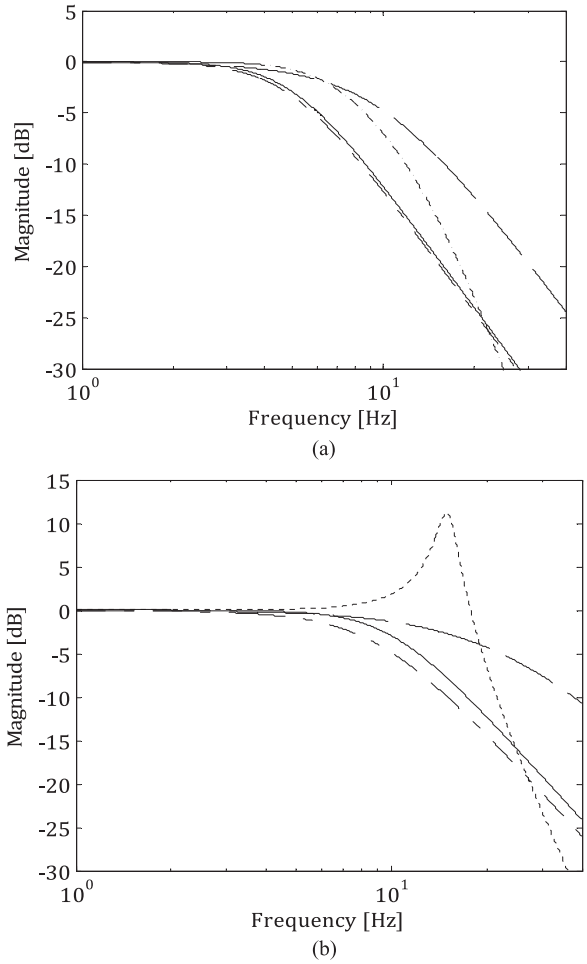


Fig. 1. Comparison of the models for the damping ratio $\zeta = 0.7$[sold: (44), dash-dot: (45), dashed: (46), dotted: (47)]. (a) $\omega_s = 5$ Hz. (b) $\omega_s = 10$ Hz.

VI. Gain Selection for the Proposed Control

The results in Section V suggest that the error of the slow model (9) with or without the corrective control (20) can be small only at low frequency by choosing small gains of the slow control. Another limitation is that the modeling error cannot be reduced by adjusting the bandwidth of the fast system as the bandwidth is given by the inertia (50), which is not the case for the proposed control as shown in (32). Furthermore, the proposed slow model (43) is more accurate than the rigid model near the dominant poles. Therefore, the proposed control offers more opportunities to reduce the modeling error compared with the conventional control.

For the proposed control, let us consider the gain,

$$\mathbf{L}_d = l_d \mathbf{I}, \mathbf{L}_p = l_p \mathbf{I}, \mathbf{K}_d = k_d \mathbf{I}, \text{and } \mathbf{K}_p = k_p \mathbf{I} \tag{52}$$

such that the slow model (43) is decoupled. Then, the system (2) with the proposed control can be written in two-time scales,

$$\ddot{\mathbf{q}}_r + \gamma_{\varepsilon,d}\, \dot{\mathbf{q}}_r + \gamma_{\varepsilon,p}\, \mathbf{q}_r = \mathbf{0} \tag{53a}$$

$$\boldsymbol{\mu}'' + l_d\, \boldsymbol{\mu}' + l_p\, \boldsymbol{\mu} = \mathbf{0}. \tag{53b}$$

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Let us assume that the desired performance of (53) is given by,

$$\ddot{\mathbf{q}}_r + 2\zeta\omega_s\dot{\mathbf{q}}_r + {\omega_s}^2\mathbf{q}_r = \mathbf{0} \tag{54a}$$

$$\boldsymbol{\mu}'' + 2\zeta\omega_f\boldsymbol{\mu}' + {\omega_f}^2\boldsymbol{\mu} = \mathbf{0}. \tag{54b}$$

In [25], it is shown that the linear system (3) is stable regardless of the perturbation parameter under (31) and (36) as long as the fast system (54b) is at least twice as fast as the slow system (54a). The bandwidth of the fast system can be written as

$$\omega_f = a\varepsilon\omega_s, a > 0. \tag{55}$$

Therefore, the stability is guaranteed as long as it satisfies $a > 2$. In addition, as implied in (47), the full model approximates the rigid model as $a \to \infty$. Since a cannot be arbitrarily large in practice, some error will be inevitable between the rigid model and the full model even with a large value of a. In fact, this "small" error can be further reduced by the proposed slow model (53a) because this model is more accurate than the rigid model.

Suppose that the parameter a has been chosen according to $a > 2$ to satisfy the stability condition. The gain $\gamma_{\varepsilon,d}$ and $\gamma_{\varepsilon,p}$ can be calculated such that the system (53a) can achieve the desired performance (54a) by the following computation:

$$k_p = \frac{a\,{\omega_s}^3}{a\,\omega_s + 2\zeta k_d} \tag{56a}$$

$$k_d = \frac{\beta^{1/3}}{6} - \frac{a\,\omega_s}{3\zeta} + \frac{a\,{\omega_s}^2\left(12\zeta^2 + a\right)}{6\beta^{1/3}\zeta^2} \tag{56b}$$

where $\beta \triangleq 6\sqrt{3\alpha} + \frac{a\,{\omega_s}^3}{\zeta^3}[a^2 + 18(a+3)\zeta^2] > 0$, and

$$\alpha \triangleq \frac{a^2{\omega_s}^6}{\xi^4}\left[\left(1-\xi^2\right)a^2 + 2\xi^2\left(9 - 8\xi^2\right)a + 27\xi^2\right] > 0.$$

It can be shown that the gain (56) is also valid for the rigid model, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x}, 0))$, i.e., $k_p > 0$ and $k_d > 0$. To see this, let us first check for k_d. It holds

$$k_d > 0 \;\Leftrightarrow\; 6\beta^{1/3}\zeta^2 k_d > 0. \tag{57}$$

Multiply $6\beta^{1/3}\zeta^2$ to (56b) to obtain

$$\begin{aligned} 6\beta^{1/3}\zeta^2 k_d &= \zeta^2\beta^{2/3} - 2\zeta a\,\omega_s\beta^{1/3} + a\,{\omega_s}^2\left(12\zeta^2 + a\right) \\ &= \zeta\beta^{1/3}(\zeta\beta^{1/3} - 2a\,\omega_s) + a^2{\omega_s}^2 + 12\zeta^2 a\,{\omega_s}^2. \end{aligned} \tag{58}$$

It is trivial to show that

$$\beta^{1/3} > \frac{a\,\omega_s}{\zeta}. \tag{59}$$

Applying (59) to (58),

$$\begin{aligned} &6\beta^{1/3}\zeta^2 k_d > a\,\omega_s\,(a\,\omega_s - 2a\,\omega_s) + a^2{\omega_s}^2 + 12\zeta^2 a\,{\omega_s}^2 \\ &\to 6\beta^{1/3}\zeta^2 k_d > 12\zeta^2 a\,{\omega_s}^2 \\ &\to 6\beta^{1/3}\zeta^2 k_d > 0 \\ &\to k_d > 0. \end{aligned} \tag{60}$$

By (56a), it holds $k_d > 0 \to \; k_p > 0$.

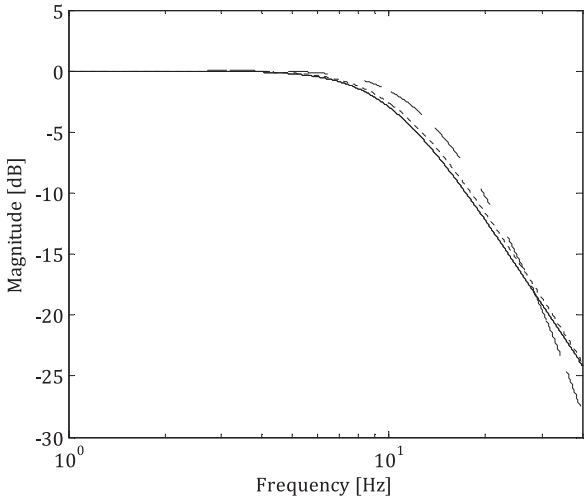


Fig. 2. Comparison of the models for the damping ratio $\zeta = 0.7$ and $\omega_s = 10$ Hz (sold: (54a), dashed: full model with $a = 4$, dotted: full model with $a = 8$).



Fig. 3. HA006B (Hyundai industrial robot, DOF: 6, maximum payload: 6 kg, main applications: handling and arc welding).

In Fig. 2, the desired performance of the slow system in the proposed control is compared with the full model with (55) and (56) for the robot used in Section V. The modeling error is much reduced compared with the conventional scheme.

VII. Simulation and Experiment

Simulation and real-world experiment were carried out with a commercial industrial robot (see Fig. 3). The kinematic data of the robot can be found at http://www.hyundai-robotics.com/www/english/product/product1.html.

The inertia, the center of mass, and reduction ratio were obtained from the CAD data. To find the values of the joint stiffness, a multivariable frequency response function (FRF) was used based on the linear model (2). However, the model (2) is not adequate for parameter identification as the friction was not included (it is assumed that the friction can be canceled by feedforward control). Therefore, for parameter identification the following model was used assuming that the friction was dominant on the motor side:

$$\mathbf{M}\,\ddot{\mathbf{x}} + \boldsymbol{\Lambda}\dot{\mathbf{x}} + \boldsymbol{\Gamma}\,\mathbf{x} = \mathbf{f} \in \Re^{2n} \tag{61}$$

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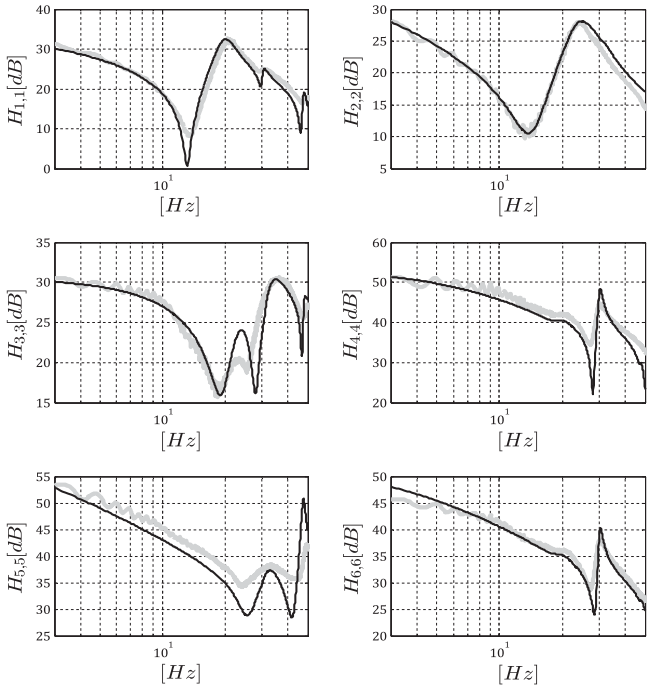


Fig. 4. Diagonal elements of multivariable FRF tested at the home position, $\mathbf{H}(\omega)_{i,j}$ (input: motor torque of joint *j*, output: motor velocity of joint *i*, gray lines: measured FRF, black lines: estimated FRF).

where $\mathbf{x} \triangleq [\mathbf{q}_r \ \mathbf{q}_m]^T$, $\mathbf{M} \triangleq \begin{bmatrix} \mathbf{M}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_m \end{bmatrix}$, $\mathbf{\Lambda} \triangleq \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{f}_v \end{bmatrix}$, $\mathbf{f}_v$: viscous friction, $\mathbf{\Gamma} \triangleq \begin{bmatrix} \mathbf{K} & -\mathbf{r}^{-1}\mathbf{K} \\ -\mathbf{r}^{-1}\mathbf{K} & \mathbf{r}^{-2}\mathbf{K} \end{bmatrix}$, and $\mathbf{f} \triangleq [\mathbf{0} \ \mathbf{u}]^T$. The model (61) can be represented in the frequency domain as follows:

$$\mathbf{x}(\omega) = \mathbf{G}(\omega)\mathbf{f}(\omega) \in \mathbb{C}^{2n} \tag{62}$$

where $\mathbf{G}(\omega) \triangleq (j\omega\,\mathbf{\Lambda} + \mathbf{\Gamma} - \omega^2\mathbf{M})^{-1}$. For a typical industrial robot such as the robot shown in Fig. 3, the measurement is available only on the motor side, namely, motor position and motor torque. Thus, the transfer function between the motor velocity, $\dot{\mathbf{q}}_m$, and the motor torque, $\mathbf{u}$, was identified by measurement,

$$\dot{\mathbf{q}}_m(\omega) = \mathbf{H}(\omega)\,\mathbf{u}(\omega) \in \mathbb{C}^n \tag{63}$$

where $\mathbf{H}(\omega) \triangleq j\omega\,\mathbf{G}(\omega)_{n+1\cdots 2n,\,n+1\cdots 2n} \in \mathbb{C}^{n\times n}$. The motor velocity was estimated by numerical differentiation from the measured motor position. A more detailed method on parameter identification of industrial robots can be found in [1]. In Fig. 4, the FRFs are shown. The lowest natural frequency at the tested robot configuration is about 10 Hz.

In Fig. 5, the block diagram of the proposed control is shown. For feedforward control, the method proposed in [10] was used. The feedforward control requires the fourth derivative of the desired link position to reproduce the nominal input. The desired trajectory was the second-order (trapezoidal velocity profile) time-optimal trajectory [26] combined with a double simple moving average filter (filter size: 80 ms = 50 ms + 30 ms). The purpose of the filter is to limit the bandwidth of the nominal trajectory as well as to increase the order of the trajectory from two to four. In simulation, the single link robot used in Sections V and VI was used. The gain of the conventional control was

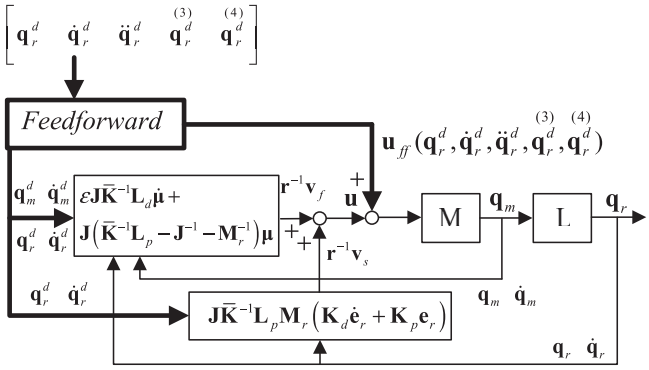


Fig. 5. Block diagram for the proposed control (M: motor side, L: link side, thick lines: feedforward paths, think lines: feedback paths, $\mathbf{e}_r \triangleq \mathbf{q}_r^d - \mathbf{q}_r$).

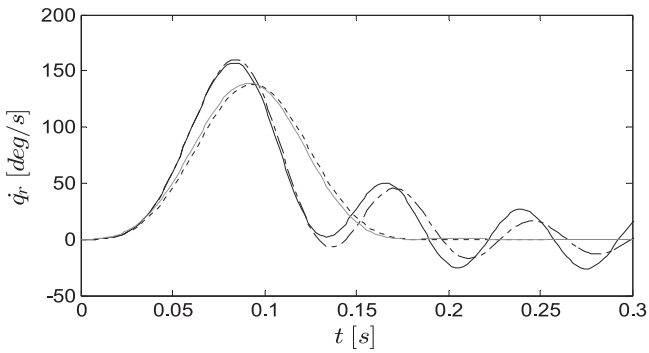


Fig. 6. Matlab simulation for a single link robot for $\zeta = 0.7$ and $\omega_s = 10$ Hz (dotted: reference, sold: conventional without corrective control, dash-dot: conventional with corrective control, gray: proposed control, $a = 4$).

computed according to (50) and (51). For the proposed control, the gain was calculated based on (55) and (56). In Fig. 6, the result of the simulation is shown. As expected in Fig. 1, the conventional control obtained a poor performance due to the large modeling error and the corrective control did not improve the performance significantly. The performance was much improved in the proposed control due to the smaller modeling error as expected in Fig. 2.

For real-world experiment, the conventional and proposed SPC were implemented in the Hyundai Hi5a controller (http://www.hyundai-robotics.com/www/english/product/product1.html) and were tested with the robot shown in Fig 3. Since the robot had no measurement on the link side, the link states, $\mathbf{q}_r$ and $\dot{\mathbf{q}}_r$, were estimated by a nonlinear observer [27]. The sampling rate of the control loop was 2 ms. The conventional SPC became unstable as soon as the motor was engaged unless a very small gain was selected. It was not possible to find a suitable strategy for the conventional SPC such that it does not exhibit instability with a proper gain. Therefore, the conventional SPC was excluded from further experiment for a safety reason. The proposed method was able to stabilize the system up to the closed-loop bandwidth, 10 Hz of the slow system. However, the proposed control exhibited a chattering

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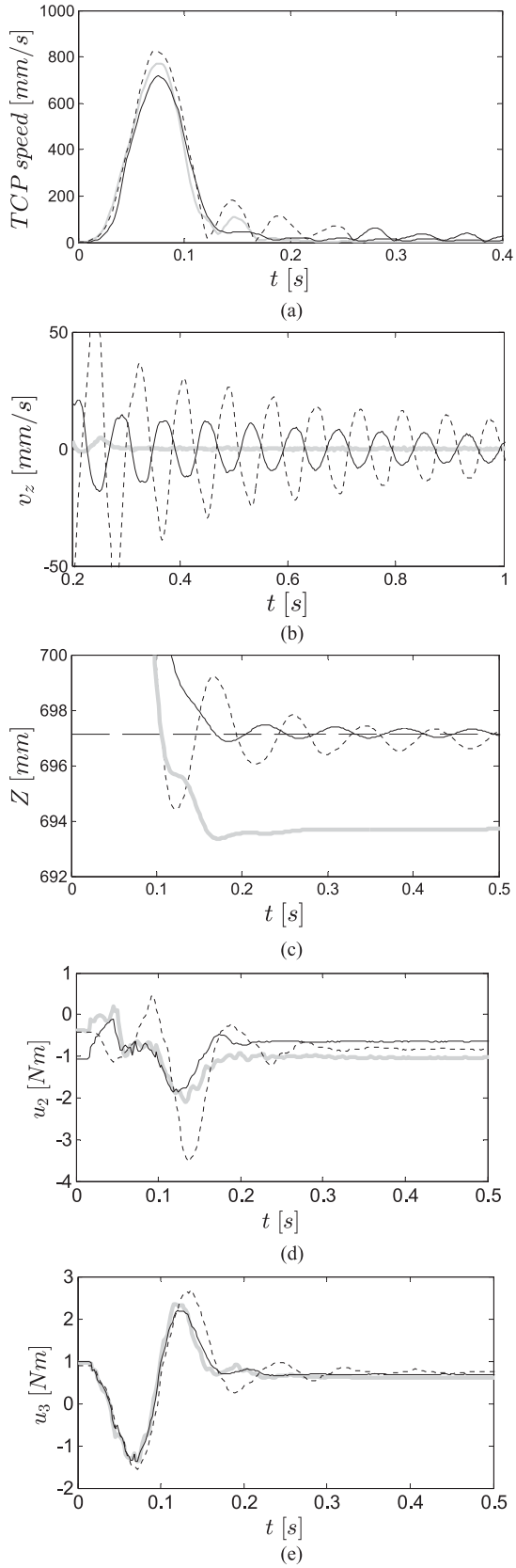


Fig 7. Experimental results for a six DOF robot (dotted: rigid controller, black-sold: feedforward controller, thick gray: proposed controller). (a) TCP speed. (b) TCP velocity in the vertical direction. (c) Steady state error in the vertical direction (dashed line: desired). (d) Motor torque of the second joint. (e) Motor torque of the third joint.

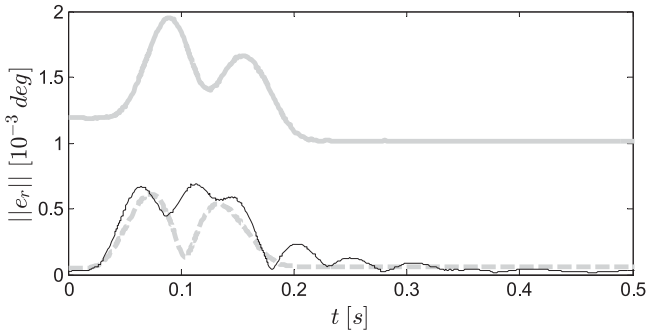


Fig. 8. Tracking errors on the link side in the HIL simulation (black-solid: feedforward control, gray-solid: proposed control at 10 Hz, gray-dotted: proposed control at 20 Hz).

phenomenon if the bandwidth was larger than 10 Hz. The gain of the nonlinear observer was also tuned at 10 Hz. Two additional model-based controllers, the computed torque controller (rigid controller) [24] and the feedforward controller [10], were also tested for comparison. The feedback control of these controllers was realized by the default PD controller of the robot (a cascaded PD controller on the motor side with sampling rate 0.4 ms). The test motion was generated such that the tool-center-point (TCP) of the robot was moved along a straight line path [length: 50 mm, downward from its home position (Fig. 3)]. A three-dimensional laser tracker, Tracker3 was used to measure the TCP position. The specification of the measurement system can be found at http://apitechnical.com/Downloads/2012/T3-Product-Specifications.pdf.

The trajectory was similar to the one used in the simulation (a second-order time-optimal trajectory with a low-pass filter). The test motion was chosen to excite the second and third links simultaneously. The dynamic coupling between the second and the third links (shoulder and elbow) is significant. In Fig. 7, the experimental results are shown. The rigid controller obtained a poor performance, generating a large path error with significant residual vibrations. The performance was much improved in the feedforward control. In the proposed control, the residual vibrations were further improved [see Fig. 7(b)]. This result verifies the effectiveness of the proposed control at steady state. That is, a faster active damping can be achieved by the proposed multivariable control (centralized full-state control) compared with the cascaded PD control (decentralized partial-state control). However, the tracking error of the proposed control at steady state was much larger than the PD control [see Fig. 7(c)]. The reason for the large tracking error is due to the lower bandwidth of the control loop compared with the PD control, especially at the second joint [see Fig. 7(d)]. The inner loop of the PD control for industrial robots is generally very high, typically varying between 80 and 100 Hz. The bandwidth of the proposed control in the experiment is only 10 and 40 Hz for the slow and fast system, respectively. Due to the limited sampling rate (2 ms) of the given controller (implementation on the motor side (0.4 ms) was not possible due to hardware limitation), it was difficult to increase the gain further [28]. Therefore, the proposed control was tested in a hardware-in-the-loop (HIL) simulation to investigate whether the tracking error could be reduced by increasing

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the gain. In the simulation, the sampling period was reduced to 1 ms. In Fig. 8, the simulation result is shown. As in the experiment, the proposed control generated a large tracking error at the bandwidth with 10 Hz. The tracking performance finally became similar to the feedforward control when the bandwidth of the slow system reached 20 Hz. If the sampling period was 2 ms, the same gain could not be used as it generated a chattering phenomenon (in simulation). This implies that the sampling period of the proposed control should be at least 1 ms to achieve a similar tracking performance as the feedforward control.

VIII. Conclusion

This paper has presented a novel two-time scale controller for flexible joint robots. First, the limitations of the conventional two-time scale controller were discussed. It was shown that the most significant problem of the conventional method is a large modeling error of the slow model that cannot be compensated by a certain corrective control. The reason why such corrective control cannot improve the modeling error was clearly explained based on a linear model. Based on the analysis on the conventional method, a more accurate slow model has been proposed by an explicit computation of the invariant manifold. A method was also presented to compute the gain such that the modeling error of the two-time scale control could be very small regardless of the perturbation parameter. Finally, the simulation and experimental results showed that the proposed controller could be more effective to improve the motion performance of flexible joint robots compared with some of model-based PD controllers.

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